Research Plan: Verification, Monitoring & Enforcement of AI Pacing

This is partially generated with AI, and reviewed in detail.

Objective

Develop a foundational understanding of *pacing the AI frontier*, with a particular focus on mechanisms for independently verifying, monitoring, and enforcing agreements to slow or constrain frontier AI capability development.

We are interested not only in what is technically feasible, but also in the **invasiveness, incentives, trade-offs, and unintended consequences** of different approaches. The aim of this plan is to build a shared understanding of the landscape, rather than reach conclusions.

Feel free to use LLMs to summarise the reading, or ask it specific questions to further your understanding. Understand that the pacing literature is very new, with pacing.tech being a often referred-to source.

1. Understanding AI Pacing

Core reading

- Pacing the Frontier: A Framework and Research Agenda — Douglas et al. (2026). The central conceptual framework. Focus on the rationale for pacing, what might be paced, possible interventions, and implementation challenges.
- Pacing the Frontier — Employee Statement (2026). A concise statement of the case for building internationally coordinated pacing mechanisms.

Consider how pacing differs from earlier proposals to pause or stop AI development, and the distinction between slowing *capability development* (slowing models getting better) and slowing *deployment or diffusion* (slowing how fast capability of deployed models improve, i.e. what everyday people experience)

2. Frontier Lab Perspectives

Understanding frontier labs' views is particularly important: any successful mechanism needs to account for their commercial incentives, security concerns, and willingness to cooperate.

Core reading

- We Must Pace the Frontier — Dario Amodei (2026). A frontier-lab perspective on the need for pacing, international coordination, and independent oversight.
- Priorities and Principles for Effective Third-Party Assessments — OpenAI (2026). Discusses independent evaluation, access to internal systems, confidentiality, and protections for sensitive intellectual property.

As you read, consider what labs appear willing to accept voluntarily, where their incentives might diverge from effective oversight, and whether different frontier labs have materially different positions.

3. Verification, Monitoring & Enforcement Mechanisms

Distinguish three functions (these aren't all fully established terminology, but a good way to think about mechanisms):

- **Verification:** Independently establishing whether a lab has complied with an agreed restriction.
- **Monitoring:** Continuously observing relevant activities, resources, or capabilities.
- **Enforcement:** Technically restricting activity without requiring the lab to cooperate at the moment of intervention, possibly following prior agreement to the mechanism or some government mandated intervention.

Core reading

- Verification Methods for International AI Agreements — Wasil et al. (2024). A broad taxonomy of technical verification methods.
- Verifying International Agreements on AI: Six Layers of Verification — Baker et al. (2025). Explores complementary approaches to achieving credible verification.

Optional technical reading

- Hardware-Level Governance of AI Compute — Ansari (2026). A technical assessment of compute monitoring, verification, and enforcement.
- Flexible Hardware-Enabled Guarantees for AI Compute — Petrie et al. (2025). A proposal for hardware-based governance mechanisms.

Five initial mechanism families worth understanding are **independent third-party audits, cloud/infrastructure monitoring, hardware-based controls, cryptographic verification, and research workflow/model lifecycle controls**. These are overlapping approaches rather than mutually exclusive alternatives.

Independent third-party audits: External auditors assess a lab's compliance through access to internal records, systems, personnel, and potentially model artefacts.

Cloud/infrastructure monitoring: Cloud providers or datacentre operators independently track compute usage and workloads, potentially restricting access when agreed limits are reached.

Hardware-based controls: Security features embedded in chips or associated hardware independently verify resource usage and potentially prevent unauthorised computation.

Cryptographic verification: Use cryptographic techniques to prove compliance with specified restrictions without revealing sensitive proprietary information.

Research workflow/model lifecycle controls: Monitor and gate activities such as AI research experiments, model training, checkpoint creation, and deployment through independently controlled permissions.

For each, consider what it can actually verify or enforce, which parties need to cooperate, how readily it could be circumvented, and whether it meaningfully slows capability development rather than merely satisfying a measurable restriction.

4. Trade-offs, Incentives & Unintended Consequences

A central question is whether stronger verification and enforcement necessarily produce better outcomes, given the costs and behavioural responses they may generate.

Particularly important considerations include:

- **Espionage and confidentiality:** Exposure of commercially sensitive research, model weights, intellectual property, or national-security information.
- **Participation and buy-in:** Whether invasive requirements make labs or infrastructure providers unwilling to cooperate.
- **Evasion and covert development:** Incentives to relocate research, use alternative infrastructure, or conceal activity.
- **Performative compliance:** Satisfying the letter of a restriction while continuing to accelerate capabilities through other means.
- **Implementation costs:** Operational disruption, compliance costs, technical complexity, and effects on beneficial or safety research.
- **Concentration of power:** Giving governments, auditors, cloud providers, or chip manufacturers excessive influence over frontier AI development.

It may be useful to distinguish three forms of *invasiveness*: access to sensitive information, interference with normal operations, and surrender of decision-making authority.

An important overarching question is whether a less intrusive mechanism that attracts widespread participation could outperform a stronger mechanism that incentivises resistance or covert development.

5. Questions to consider

1. Which mechanisms seem most promising for credible, independently verifiable pacing, and why?
2. Which controls target capability development itself, rather than simply compute expenditure or model deployment?
3. Where are the largest gaps between theoretical mechanisms and practical implementation?
4. How might frontier labs, cloud providers, and governments respond strategically to different requirements?
5. Can we combine mechanisms to obtain strong assurance with comparatively limited invasiveness?
6. What are the most important questions or approaches missing from the existing literature?

Bonus economics focused questions

7. Incentive compatibility: Under what conditions would frontier labs voluntarily participate in pacing agreements, comply with restrictions, and continue cooperating as their competitive incentives change?
8. Strategic responses and perverse incentives: How might labs adapt to different restrictions through regulatory arbitrage, covert development, or substituting toward unrestricted approaches that continue accelerating capabilities?
9. Information asymmetries and principal–agent problems: How do differences in information and incentives between labs, auditors, cloud providers, and regulators affect the credibility of verification and enforcement?
10. Market structure and competition: Could verification requirements inadvertently advantage incumbent labs, increase barriers to entry, concentrate market power, or alter competitive dynamics between firms and countries?
11. Optimal mechanism design: How should we balance verification strength, invasiveness, compliance costs, and willingness to participate? Could less intrusive mechanisms achieve better overall pacing outcomes through broader adoption?

Optional LLM Prompts

Understanding the technical landscape

"Compare the principal methods for independently verifying, monitoring, and enforcing frontier AI pacing agreements. Explain their technical foundations, maturity, what they can and cannot establish, and how they might be combined. Distinguish existing deployments from theoretical proposals, and provide primary sources."

Exploring incentives and unintended consequences

"Analyse the incentives created by different AI pacing verification mechanisms for frontier labs, cloud providers, governments, and independent auditors. Consider information asymmetries, principal-agent problems, Goodhart's law, regulatory arbitrage, covert development, and the trade-off between invasiveness and willingness to participate."

The goal is to come to our next discussion with a shared understanding of the main ideas, practical constraints, and open questions — not necessarily settled views on which mechanisms to pursue.